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Range-wide genetic structure and diversity of the endemic tree line species *Polylepis australis* (Rosaceae) in Argentina[1]

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- *Premise of the study:* Knowledge on the range-wide distribution of genetic structure and diversity is required to facilitate the understanding of historical tree migration and for predicting responses to current climate change. With respect to post-glacial migration patterns known from the northern hemisphere, we tested the prediction that the southernmost populations of a subtropical tree line species have lower within-population genetic diversity and higher genetic differentiation than the central and northernmost populations.
- *Methods:* We used AFLP to assess the genetic structure of 18 populations of the wind-pollinated *Polylepis australis* (Rosaceae) sampled over its entire distributional range in three Argentinean high mountain regions. Genetic diversity was calculated as a percentage of polymorphic bands (P) and Nei's expected heterozygosity (He); genetic differentiation was assessed using AMOVA, Φ_{ST} – statistics, and Bayesian cluster analysis.
- *Key results:* Contrary to our expectations, the northernmost *Polylepis australis* stands had lower within-population genetic diversity and higher genetic differentiation than the central and southernmost stands. Populations grouped into two major clusters, the first containing the southern populations and four central populations and the second containing the northern and one central population.
- *Conclusions:* Patterns of *Polylepis australis* genetic structure and diversity differ from historical migration scenarios observed for the northern hemisphere. The decline in genetic diversity toward the north may point to an equatorward migration following past climatic changes. Populations within the south and central part appear to be connected by effective long-distance pollination while gene flow in the northern part is probably hampered by geographic isolation.

Key words: AFLP; climate change; high mountain forests; neotropics; population genetics; tree migration

Data on the range-wide genetic structure and diversity of dominant tree species facilitates the estimation of long-term responses to climate change in regions where fossil pollen records are lacking. The combined influence of evolutionary, environmental, and anthropogenic factors acting at different time scales is reflected in such studies (Victory et al., 2006). In this respect, genetic structure was highly impacted by climatic fluctuations during the Quaternary Period (Hewitt, 2000; Hu et al., 2008). Such knowledge, along with that of the current distribution of the genetic diversity of dominant tree species, facilitates our understanding of phylogeographical patterns, including those of postglacial migrations (Taberlet et al., 1998; Hewitt, 2000), allowing for the prediction of future responses to climate change.

In the northern hemisphere, genetic diversity of tree species frequently has been shown to decrease with latitude due to refugia located in warmer equatorward regions, long-range postglacial migration patterns, and successive founder events (Hewitt, 2000; Petit et al., 2003). In the southern hemisphere, knowledge of genetic structure patterns and their implications for tree migration is still scarce, in particular for high mountain species (Quiroga and Premoli, 2007; Pautasso, 2009). The few genetic studies on tree migration in the South American Andes failed to reveal a similar scenario to that described for the northern hemisphere. Declines in genetic diversity to the north and with higher elevation found for two South American Podocarpaceae (*Podocarpus parlatorei* Pilg. and *P. nubigenus* Lindl.) suggest that cold-tolerant conifer forests, in response to the Quaternary glaciations and interglacial periods, retreated to the north and to lower elevations during cooler climates and migrated to higher elevations during warmer periods (Quiroga and Premoli, 2007; 2010). Other scenarios indicate that mountain areas, due to their heterogeneous topographies, could harbor multiple glacial forest refugia, as genetically shown for *Araucaria araucana* (Molina) C. Koch, *Fitzroya cupressoides* (Molina) I.M. Johnst., several *Nothofagus* species in temperate South America (Marchelli et al., 1998; Premoli et al., 2000; Bekessy et al., 2002; Mathiasen and Premoli, 2010), and for

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Juniperus tibetica Kom. in the southern Tibetan Plateau (Opgenoorth et al., 2010).

The genus *Polylepis* (Rosaceae) encompasses between 15 and 30 wind-pollinated tree and shrub species that are distributed from southern Venezuela to central Argentina (Schmidt-Lebuhn et al., 2010). Some species occur in the area of the (sub) tropical upper mountain forests above the vegetation belt colonized by *Podocarpus* species, while others develop isolated stands far above the closed tree line up to altitudes of 5200 m a.s.l., where they form the world's highest forests. This distribution pattern is probably the result of deforestation of formerly extensive forests (Renison et al., 2006; Cingolani et al., 2008), but historical climatic changes may also have played a role (Di Pascuale et al., 2008; Gosling et al., 2009). Presently, the *Polylepis* forests of the Andean highlands represent one of South America's most endangered forest ecosystems. P. *australis* Bitt. is the southernmost species of this genus and is endemic to Argentina. Its distributional range extends to about 1200 km from the central Argentinean Córdoba Mountains to the Andes in Northern Argentina, almost reaching the border with Bolivia (Fig. 1). About 90% of the distribution of *P. australis* lies between 1500 and 3500 m a.s.l., with single individuals found down slope as low as 900 m a.s.l. in the Córdoba Mountains (Marcora et al., 2008). Its current distribution is restricted to three large mountain ranges separated by extensive dry lowlands at ca. 400 m a.s.l. for the southern and central ranges, and at ca. 800 m a.s.l. for the central and northern ranges. Previous studies on the mating system of *P. australis* revealed high pollen viability and longevity, effective pollen flow, and that selfing is hampered by protogyny (Seltmann et al., 2007; 2009). Fruits are single-seeded nutlets that, at least in the southern distributional range, are produced annually and dispersed short distances by wind (up to 6 m, Torres et al., 2008; Zimmermann et al., 2009).

To determine effects of repeated phases of range contraction and expansion during the Quaternary ice ages, we analyzed genetic structure and diversity of *Polylepis australis* populations along its entire distributional area, using amplified fragment length polymorphisms (AFLP). We specifically asked: 1) whether genetic diversity and genetic differentiation differ among and within the three extant distributional areas; and 2) whether genetic characteristics allow us to draw preliminary conclusions on historical forest migration patterns. Our underlying hypothesis is that the current genetic structure of populations of the cold-tolerant *P. australis* reflects a long-term response to climate changes, such as those that occurred during the Quaternary. Under a scenario in accordance with the large-scale glaciations and migrations known from the northern hemisphere, we expected that genetic diversity of *P. australis* would decrease with decreasing latitude (poleward). Under an

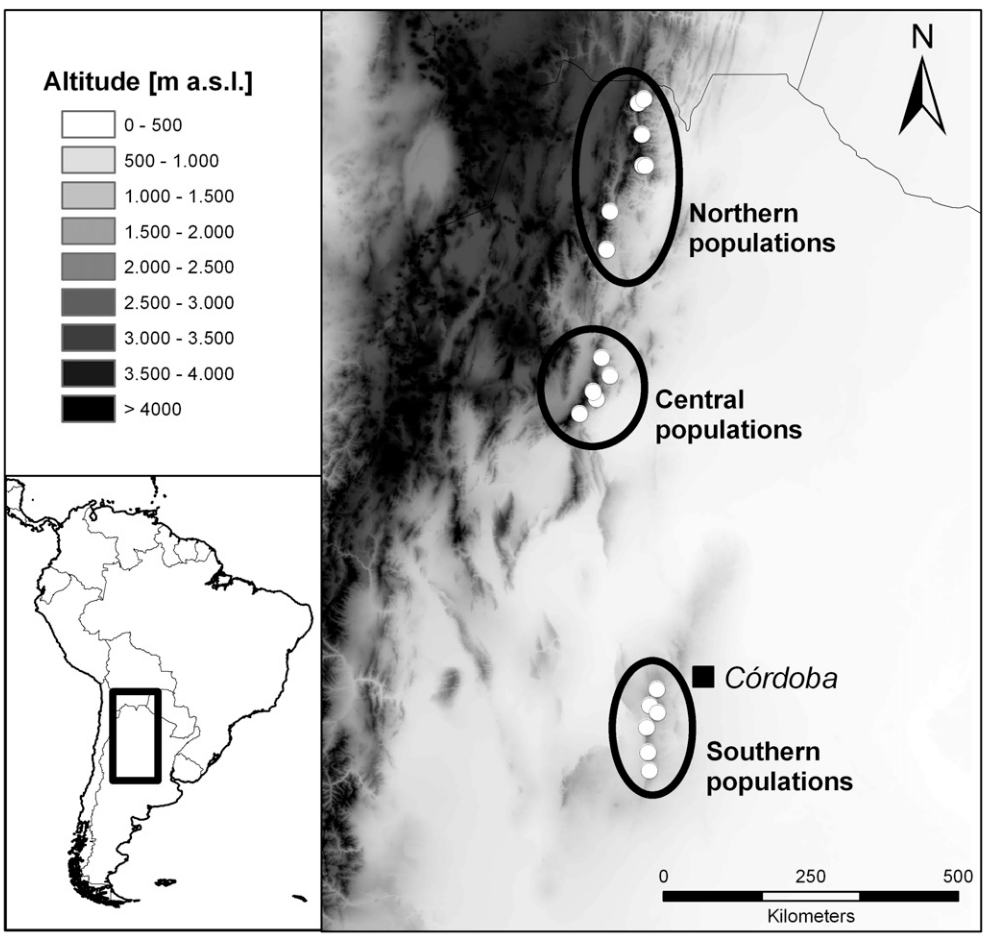

Fig. 1. Location of the sampled populations of *Polylepis australis* endemic to Argentina. Sampling is comprehensive, as gaps between the northern, central and southern populations (shown with ellipses) are not colonized by *P. australis*. Population numbers refer to Table 1.

alternative scenario of modest glaciations, we might expect a pattern of increasing genetic diversity with decreasing latitude due to successive downward and northward migrations during cold periods and upward movements during warmer periods, as found for two cold-tolerant *Podocarpus* species (Quiroga and Premoli, 2007; 2010). Additionally, we expected that the highest elevation populations may be genetically impoverished due to climate driven shifts in elevation and successive founder events. This study represents the first population genetic study on a *Polylepis* tree line species covering its entire distributional range.

MATERIALS AND METHODS

Study sites and sampling design—We collected leaf material from a total of 18 populations in three mountain ranges covering the whole distributional area of *Polylepis australis*. The southernmost range was situated in the Córdoba Mountains (7 populations; 1716-2002 m a.s.l.), the central range in the Aconquija Mountains (5 populations; 2040-2738 m a.s.l.) and the northern range in the eastern Andes of northwestern Argentina (6 populations; 2404-3416 m a.s.l.); hereafter referred to as southern, central and northern populations, respectively (Fig. 1, Table 1). In the three study regions, populations were sampled at similar spatial scales, i.e., the minimum and maximum distances between the northern populations were 5 km and 264 km respectively, between the central populations, 13 km and 101 km, and between the southern populations, 17 km and 139 km. Areas between regions were devoid of any *Polylepis* trees and spanned 184 km (northern vs. central populations) and 482 km (central vs. southern populations), respectively. Within each forest, we randomly sampled 13 individuals. Sizes of the sampled areas varied because of differing densities and population sizes (Table 1; mean of 96 adult individuals/ha, Renison et al., 2011), but there were more than 1000 individuals in all but two of the locations (Merlo and Mina Clavero populations in the southern region each with ca. 500 individuals). Sampled trees were separated by at least 200 m to minimize the chance of sampling closely related or genetically identical individuals. The total sample size successfully analyzed comprised 208 individuals (between 8 and 13 samples per population; mean 11.6). Leaves were stored in bags with silica-gel prior to analysis.

DNA extraction and AFLP analysis—The extraction method followed a standard protocol adapted from Doyle and Doyle (1987) using 20 mg silica-gel-dried leaf material and a modified extraction buffer [2% alkyltrimethylammonium bromide (ATMAB), 0.1 M Tris- HCl, 0.02 M disodium-EDTA (pH 8.0), 1.4 M NaCl, 1% PVP]. Extracted genomic DNA was double-digested with the restriction enzymes MseI and EcoRI, and the ends of the resulting fragments were ligated to double-stranded adapter oligonucleotides (5-GACGATGAGTCCTGAG-3 / 5-TACTCAGGACTCAT-3 and 5-CTCGTAGACTGCG-TACC-3 / 5-AATTGGTACGCAGTCTAC-3) serving as primer binding sites in the following steps. Restriction and ligation were performed for three hours at 37°C, followed by 10 min at 65°C in an 11 µl volume containing 1 U of MseI, 5 U of EcoRI, 1 U of T4 DNA ligase, 1.1 µl T4 DNA ligase 10× reaction buffer (all New England Biolabs, Frankfurt am Main, Germany), 0.05 mM NaCl, 0.05 mg/ml BSA, 5 pmol of EcoRI-adapter, 50 pmol MseI adapter, and 5.0 µl DNA extract. The ligation product was diluted with 39 µl of sterile demineralised water and then preamplified with the primer combination EcoRI+A / MseI+C (E01, 5-GACTGCGTACCAATTC+A-3 / M02, 5-GATGAGTCCTGAGTAA+C-3; primer nomenclature following KeyGene, Inc. (2004). Preamplification was performed in a 20 µl volume containing 0.5 U BioTaq DNA Polymerase, 2.0 µl PCR 10× reaction buffer, 1.5 mM $MgCl_2$, 0.2 mM of each dNTP (all Bioline, Luckenwalde, Germany), 5 pmol of both preprimers, and 4 µl of the ligation product with the following temperature profile: 5 min initial denaturation at 94°C, 20 cycles of 20 s denaturation at 94°C, 30 s annealing at 56°C, and 120 s elongation at 72°C.

The preamplification product was diluted 10-fold with sterile demineralised water. Selective amplification was carried out in a 20 µl volume containing 0.5 U BioTaq DNA Polymerase, 2.0 µl PCR 10× reaction buffer, 1.5 mM $MgCl_2$, 0.2 mM of each dNTP (all Bioline, Luckenwalde, Germany), 5 pmol MseI selective primer, 1 pmol fluorescence labeled EcoRI selective primer, and 3 µl preamplification product with the following temperature profile: 1 min initial denaturation at 95°C, 10 cycles of 20 s denaturation at 94°C, 30 s annealing at 65°C (decreasing by 1°C per cycle), 120 s elongation at 72°C, followed by 20 cycles of 20 s denaturation at 94°C, 30 s annealing at 56°C, and 120 s elongation at 72°C (increasing by 4 s per cycle). For the selective amplification, 21 different primer combinations were tested on a small number of samples for their level of variability within and among species, and four primer combinations were chosen for fingerprinting all samples. The four combinations were EcoRI+AAC*FAM, 5-GACTG-CGTACCAATTC+AAC-3 / MseI+CAC, 5-GATGAGTCCT-GAGTAA+CAC-3, EcoRI+AGC*HEX; 5-GACTGCGTACC-AATTC+AGC-3 / MseI+CAC, 5-GATGAGTCCTGAGT-AA+CAC-3; EcoRI+AAG*FAM, 5-GACTGCGTACCAAT-TC+AAG-3 / MseI+CTG, 5-GATGAGTCCTGAGTAA+CTC-3, and EcoRI+ACT*HEX 5-GACTGCGTACCAATTC +ACT-3 / MseI+CTG, 5-GATGAGTCCTGAGTAA+CTG-3.

AFLP main amplification products in plates (96-well plates, ABgene, Epsom, United Kingdom) were purified by centrifugation (910×g at 4°C) through Multi Screen 96-well plates (Millipore MSHVN4510, Schwalbach, Germany) on a column of Sephadex G-50 Superfine powder (GE Healthcare Bio-Science, Uppsala, Sweden), and purified amplification products were analyzed using a MegaBACE 1000 sequencer (Amersham Biosciences, Freiburg, Germany).

Data analysis—Polymorphic DNA bands were scored as present (1) or absent (0) for each DNA sample. Smeared and weak bands were excluded via visual inspection. Randomly chosen samples were used to test the reproducibility of each of the steps of the AFLP procedure, including band scoring. The four primer combinations yielded 269 reliable bands, of which 244 bands were polymorphic (90.7%) that were consequently used in the analysis. The number of polymorphic bands per primer pair of *Polylepis australis* ranged between 33 and 92.

The percentage of polymorphic bands (P), and Nei's expected heterozygosity (i.e., genetic diversity, He) were calculated to estimate the level of genetic diversity (GenAlEx 6.2, Peakall and Smouse, 2006). An analysis of molecular variance (AMOVA) was used to describe genetic structure and to measure the amount of variation found within and between populations; Φ statistics (analogs of F statistics) were extracted and significance levels were tested with 999 permutations for each analysis. Populations examined in the AMOVA procedure were initially assigned to three different groups based on their

TABLE 1. Geographical, climatic and genetic data of the sites used for leaf collection of *Polylepis australis* populations in Argentina. Populations are ordered geographically from north to south. Symbols of variables: *n* = sampling size; *Pa* = Population area; area covered with *P. australis* forests (ha); *Tc* = tree coverage; estimated percentage of population area covered by *P. australis* canopies; *Ps* = population size (Pa*Tc /100); *Lt* = latitude (in decimals); *Ln* = longitude (in decimals); *Al* = Altitude (m a.s.l.); *Rn* = mean annual rainfall (mm); *Tm* = mean annual temperature (°C); *Ta* = mean hottest month (January) temperature (°C); *Tj* = mean coldest month (January) temperature (°C); *Nb* = number of bands; *P%* = percentage of polymorphism; *He* = mean gene diversity.

					Geographical			Climatic				Genetic		
Population	*n*	*Pa*	*Tc*	*Ps*	*Lt*	*Ln*	*Al*	*Rn*	*Tm*	*Ta*	*Tj*	*Nb*	P%	He
A. Northern populations														
Vallecito	13	150	10	15	−22.22	−64.59	2863	357	11.7	24.9	−5.7	184	53.7	0.184
Portillo	12	82	20	16	−22.26	−65.04	3416	313	9.5	23.2	−8.0	177	37.3	0.127
Queñoal	13	150	10	15	−22.54	−65.01	2854	297	12.0	25.0	−4.7	184	46.7	0.155
Valle Colorado	10	250	10	25	−23.23	−64.57	2695	354	12.1	24.5	−3.7	179	40.2	0.137
Santa Ana	12	200	25	50	−23.23	−65.00	3158	258	10.4	23.2	−6.0	188	52.9	0.176
Yala	12	500	70	350	−24.05	−65.30	2624	258	10.3	22.4	−4.9	194	57.8	0.192
San Lorenzo	12	300	20	60	−24.41	−65.33	2404	318	12.5	24.1	−2.2	186	45.5	0.152
							Northern populations, mean:					184.6	47.7	0.1604
							Northern populations, variance:					31.9	55.9	0.0006
B. Central populations														
San Pedro C	13	130	15	19	−26.20	−65.30	2617	248	12.9	24.4	0	211	59.0	0.194
Arquitas	12	125	10	12	−26.36	−65.30	2040	394	13.8	25.3	1.6	194	63.5	0.215
El Mollar	8	150	20	30	−26.57	−65.42	2193	342	13.5	25.6	0.7	209	55.3	0.186
Ovejerias	13	80	70	56	−26.50	−65.45	2231	301	11.9	24.0	−0.8	221	60.7	0.195
Alisos	13	700	10	70	−27.10	−65.33	2738	261	12.2	24.7	−1.0	223	64.3	0.197
							Central populations, mean:					211.6	60.6	0.1947
							Central populations, variance:					133.8	13.2	0.0001
C. Southern populations														
Yuspe	10	130	15	19	−31.23	−64.46	1880	652	11.2	24.8	−1.9	208	57.4	0.184
Mina Clavero	11	100	3	3	−31.39	−64.53	1862	642	11.7	25.5	−1.5	213	62.3	0.197
Santa Clara	12	500	75	375	−31.45	−64.46	1937	668	10.7	24.4	−2.3	213	60.3	0.189
Los Molles	12	150	50	75	−31.58	−64.56	2002	650	11.2	25.1	−2.0	207	61.1	0.204
Merlo	12	80	3	2	−32.21	−64.54	1999	675	10.4	24.5	−2.8	206	55.7	0.176
Villa Larca	8	110	10	11	−32.37	−64.53	1716	687	11.7	25.8	−1.7	197	55.7	0.198
							Southern populations, mean:					207.3	58.7	0.1913
							Southern populations, variance:					34.7	8.2	0.0001

Table 2. Analysis of molecular variance (AMOVA) for 208 *P. australis* individuals grouped into 18 populations and separately into three regions (*P* based on 999 permutations), two regions, and for each region group.

All regions	df	Sum of squares	Variance	% Total	*P*-value
Between regions	2	432.8	2.5	9	<0.001
Among populations within regions	15	700.2	2.1	7	<0.001
Within populations	190	4368.3	23.0	84	<0.001
Northern region vs. southern/central region					
Between regions	1	321.6	2.7	10	<0.001
Among populations within regions	15	811.4	2.4	9	<0.001
Within populations	190	4368.3	23.0	82	<0.001
Northern region					
Among populations	6	377.0	3.6	15	
Within populations	77	1524.0	19.8	85	<0.001
Central region					
Among populations	4	177.7	1.6	6	
Within populations	54	1377.0	25.5	94	<0.001
Southern region					
Among populations	5	145.6	0.5	2	
Within populations	59	1467.4	24.9	98	<0.001

geographic origins (northern, central, and southern populations; Fig. 1, Table 1). To assess the level of differentiation at different spatial scales, we additionally calculated an AMOVA for each of the three regions, respectively; and for only two groups (northern vs. central/southern populations) due to the revealed genetic dissimilarity between these groups. AMOVA was performed with GenAlEx 6.2 (Peakall and Smouse, 2006). Mantel's test (Mantel, 1967), performed with the package vegan (Oksanen et al., 2009) in R version 2.10.0 (R Development Core Team, 2009), was used to examine whether the matrix of genetic differentiation among populations (pairwise Φ_{ST}–values) correlated with the matrix of geographical distances.

To infer ancestry groups within the data set, we performed a Bayesian cluster analysis in the program STRUCTURE 2.3.1 (Pritchard and Wen, 2004). We chose the admixture model with correlated allele frequencies among populations with 10 000 burn-in length periods and 500 000 Markov chain Monte Carlo repetitions in accordance to Falush et al. (2007) as recommended by Bonin et al. (2007) for dominant markers. Each run was repeated 10 times, with *K* ranging from 1 to 18. The optimum number of *K* clusters was assessed following inspection of the mean values of L(*K*), L'(*K*), L''(*K*) and Δ*K*, as suggested by Evanno et al. (2005). We created barplots based on the STRUCTURE output files with Distruct 1.1 (Rosenberg, 2004).

Spearman rank correlation coefficients were used to detect relationships between genetic indices (*P* and *He*) and latitude and altitude (Table 1, R version 2.10.0, R Development Core Team, 2009). Possible differences in genetic diversity parameters among the three regions were assessed with ANOVA and Tukey test.

RESULTS

Genetic diversity—Our data revealed high within-population diversity for *Polylepis australis*, as the average percentage values of polymorphism (*P*) and mean gene diversity (*He*) were 55.0 and 0.181, respectively (Table 1). However, with a mean *P* and *He* value of 47.7 and 0.160, the northern populations exhibited significantly lower genetic diversity than the central and southern populations (60.6 and 58.7, and 0.198 and 0.191, respectively, ANOVA, $F_{2, 15}$ = 10.65 and 8.16, $P < 0.01$ in both cases; Table 1).

Genetic diversity of *Polylepis australis* was negatively correlated with latitude (P: rS = −0.58, $P = 0.01$, He: rS = −0.58, $P = 0.01$, $N = 18$) and altitude (P: rS = −0.49, $P = 0.04$, He: rS = −0.53, $P = 0.02$). However, both variables were strongly related (rS = 0.88, $P < 0.001$, Table 1). When the three regions were analyzed separately, genetic diversity was not related to altitude ($P > 0.2$ in all cases). However, the highest population in the north (Portillo, 3416 m a.s.l., Table 1) had the lowest genetic diversity.

Genetic differentiation and geographical structure—When all *Polylepis australis* populations were combined into a single dataset, the analysis of molecular variance showed that 84% of the genetic variation was found within populations, 7% among populations within regions and 9% between regions ($P < 0.001$, Table 2). When two groups were formed (northern vs. central/southern populations), 9% of variance was found among populations within regions and 10% between regions. When the three study regions were analyzed separately, the northern region showed higher variation among populations (15%) than the central and southern regions, where the genetic variation among populations was lower (6% and 2%, respectively). The overall Φ_{ST} value was 0.165. Pairwise Φ_{ST} values among northern populations were significantly larger and more variable (mean 0.152; range 0.076–0.239) than those among central (mean 0.058; range 0–0.139) and those among southern populations (mean 0.015; range 0–0.029). Pairwise Φ_{ST} values were positively correlated with pairwise geographic distance for the whole data set ($r_M = 0.54$, $P < 0.001$) and for the northern populations ($r_M = 0.56$, $P = 0.02$). No isolation by distance pattern was found for the southern ($r_M = -0.02$) and central populations ($r_M = -0.14$, $P > 0.5$ in both cases).

The results of the Bayesian cluster analysis (Fig. 2) revealed an optimum number of $K = 2$ clusters for our dataset. These clusters corresponded to the southern and central region except for the Arquitas population, which clustered with the second group encompassing the northern populations. With increasing numbers of *K* ($K = 3$, $K = 4$), individuals within the southern and central regions remained within a single group, but northern individuals differentiated into two clusters, one of which contained the central Arquitas population (Fig. 2).

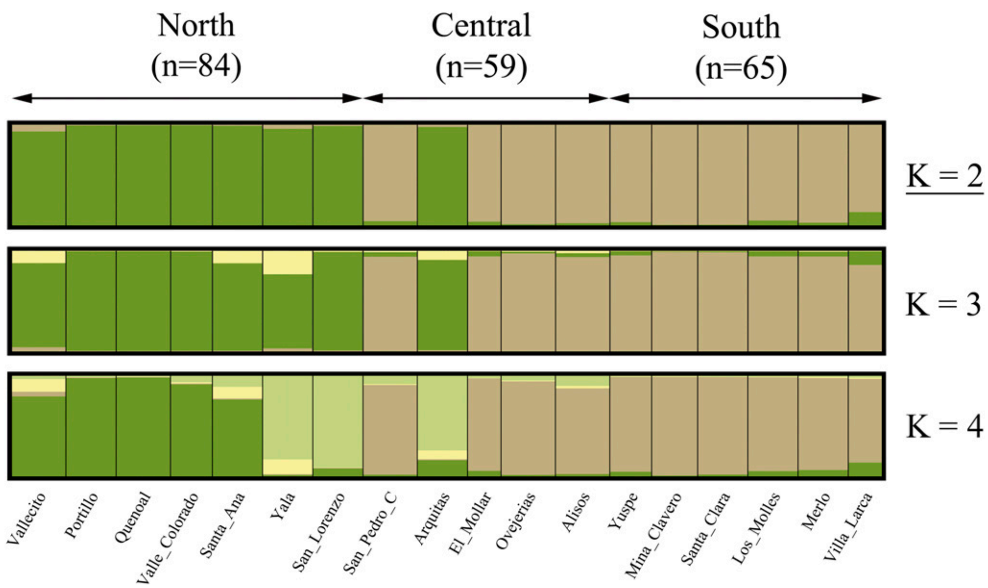

Fig. 2. Barplots showing the results of the Bayesian cluster analysis. The color in each barplot represents the probability of each individual to belong to an admixture group. The underlined cluster number $K = 2$ was found to be the optimum one. Populations are ordered geographically from north to south (in agreement with Table 1).

DISCUSSION

Genetic diversity—We found moderate to high values for genetic diversity in our studied *Polylepis australis* populations throughout the species' entire distributional range in Argentina (P = 37–64%, He = 0.127-0.215) that did not differ much from those revealed for two Ecuadorian *Polylepis* species analyzed with the same marker system (AFLP) in our laboratory: *P. pauta* Hieron. (P = 38–68%, He = 0.164-0.273, A. Cierjacks, Technische Universität, Berlin, Germany, unpublished data), and *P. incana* Kunth (P = 39–52%, He = 0.126-0.173, Hensen et al., 2011). However, our mean He value (0.181) is higher than published average AFLP data for other wind-pollinated tree species (*Abies ziyuanensis* L.K. Fu & S.L. Mo, 0.136; Tang et al., 2008; *Pinus pinaster* Aiton, 0.161; Mariette et al., 2001, *Pinus echinata* Mill. and *P. taeda* L., 0.15 and 0.12, Xu et al., 2008, and *Shorea acuminata* Dyer, 0.165, Cao et al., 2009).

Contrary to our expectation of similarity with a northern hemisphere glaciations scenario, mean genetic diversity of the northern *Polylepis australis* populations was significantly lower than that of the central and southern populations. Reduced genetic diversity is known to characterize more recently colonized populations due to founder events and bottlenecks (Hamrick, 2004; López et al., 2010). Hence, our data suggest that *P. australis* may have migrated northwards during glacial cooling and higher in elevation during Holocene Epoch warming, as recently also described for two *Podocarpus* species, the temperate *P. nubigenus* and the subtropical *P. parlatorei*. In our central and northern regions, the latter species is found as a belt below *P. australis* forests (Quiroga and Premoli, 2007; 2010). Pollen records pointing to northward and downward migration of cold-adapted tree genera such as *Podocarpus, Weinmannia*, and *Alnus* during the last glacial cycle have indeed been reported by Colinvaux et al. (1996a, b), and a high sensitivity of *Polylepis* species to past climate change was affirmed by Gosling et al. (2009), who provided the first fossil record of *Polylepis*-type pollen covering multiple glacial/interglacial cycles in the Central Andes.

However, it is remarkable also that in northern Argentina, three of seven *Polylepis australis* populations displayed elevated polymorphism and He values (Vallecito, Yala and Santa Ana, Table 1), indicating that these populations may already have persisted for a longer time. Additionally, it is remarkable in the north, that the population highest in elevation was found to have by far the lowest genetic diversity. A recent study (Hensen et al., 2011) found that genetic diversity was highly negatively correlated with altitude in both adults and seedlings of *P. incana* in Ecuador. In addition to indicating a particular colonization history or strong geographical isolation, this pattern might point to a reduced effective population size in high altitude stands due to decreasing fitness, as *P. australis* was found to have lower seed productivity and seed mass with increasing elevation (Marcora et al., 2008). For *P. incana*, a significant reduction of flower number and seedling recruitment toward the uppermost forests was also described and explained by environmental constraints, and in particular, low temperatures (Cierjacks et al., 2008).

Genetic differentiation—The distribution of genetic diversity observed in this study is in agreement with several reviews that showed that wind-pollinated tree species such as *Polylepis australis* have a high percentage of genetic variability within populations and a clearly lower variation among populations (Hamrick et al., 1992; Nybom and Bartish, 2000). Equally low fixation indices to those of our southern and central populations (<0.1) were found for several other wind-pollinated angiosperms, such as *Quercus calliprinos* Webb (Schiller et al., 2006), *Fraxinus excelsior* L. (Heuertz et al., 2001), and *Betula pendula* Roth (Rusanen et al., 2003), and in a previous study using Inter-Simple Sequence Repeat (ISSR) analysis (Julio et al., 2011).

In addition to being genetically impoverished, the northern populations of *Polylepis australis* were genetically more differentiated than central and southern populations, suggesting stronger spatial isolation and enhanced genetic drift. In accordance, Mantel's test revealed a significant correlation between genetic differentiation and geographic distance, not only for the whole data set, but also for the northern populations. Here, topography is strongly dissected, valleys are larger and steeper, and higher mountain ridges might form barriers to pollen and seed flow (Hu et al., 2008). In the central and southern regions, *P. australis* habitat is more continuous and landscape boundaries are relatively weaker (Fig. 1). This was also supported by a roughness analysis based on a digital elevation model, where the northern population revealed higher values compared to the southern and central population (H. von Wehrden, unpublished data). In fact, the results of previous studies revealed a high degree of gene flow—due to effective pollen flow—for the southern *P. australis* populations (Seltmann et al., 2007; 2009), which was hypothesized by an AFLP study by Schmidt-Lebuhn et al. (2006) for *P. rugulosa* Bitt. and *P. tarapacana* Phil. to explain a lack of genetic structure.

The Bayesian cluster analysis and AMOVA confirmed a distinct clustering of a group formed by the southern and central populations (except for the central Arquitas population), indicating high genetic similarity between them. In a second group, the central Arquitas population clustered with those of the northern study region. As this population was genetically the richest of all, and most similar to the two southernmost of the northern populations, San Lorenzo (distance 214 km) and Yala (279 km), Arquitas might have been one of the source populations for the historical northward migration of *Polylepis australis*. In comparison to the northern populations, central and southern populations showed a much more homogeneous variation of within-population genetic diversity as well as lower genetic differentiation. This result was striking as an area of almost 500 km comprised of lowlands exists between both regions. Such a pattern could arise if populations were connected in the past, not affected by bottlenecks and if there had been enough time to homogenize over long time periods. However, prevailing wind patterns in the study regions do not help explain this pattern as the region would have been situated in a transition zone between two air circulation regimes, with mainly east and west directions for most of the Pleistocene Epoch and up to the present day (Haselton et al., 2002).

Pollen data from the Northern Andes indicate that many cold-tolerant mountain species maintained populations during the glacial periods in the lowlands (Colinvaux et al., 1996a, Behling et al., 1999), and intermingled with species of the modern lowland forests. During Holocene warming, species composition of these ancient plant communities of the Andean slopes changed as heat-intolerant species ascended to their current high elevation distribution (Colinvaux et al., 1997). Our data support the premise that past *Polylepis australis* distribution included the lowland area between the central and southern regions (see Fig. 1). In concordance with genetic evidence, Haselton et al. (2002) provided data indicating a Pleistocene snowline depression of at least 900 m for our central study area, suggesting the possibility of historically increased precipitation. Remarkably, present-day dry valleys flow into salt flats now situated between the two mountain ranges, suggesting that historically, these valleys hosted rivers that may have transported *Polylepis* seeds, a dispersal mode often observed in our southernmost study region (D. Renison, personal observation). We therefore hypothesize that formerly lower temperatures, increased precipitation, and downstream dispersal of seeds may have helped to connect the *P. australis* populations in the central and southern ranges. The distributional ranges of other *Polylepis* species from Peru and Ecuador were also much more extensive during earlier periods of the Holocene than they are today (Colinvaux et al., 1997; Hansen et al., 2003; Paduano et al., 2003; Bush et al., 2005; Hanselman et al., 2005). Given present-day climate warming, a future upslope migration and range contraction of *P. australis* forests can thus be predicted.

Conclusions—The results of our study indicate that postglacial tree migration scenarios of the northern hemisphere do not coincide with those of the southern hemisphere (Premoli et al., 2007, Quiroga and Premoli, 2007; 2010), probably due to different landscape structures, a greater oceanic influence prevailing in the southern hemisphere, and the overall shorter glacial/interglacial cycles (Markgraf et al., 1995). The identified differences in genetic diversity and structure between the two genetically differentiated population groups indicate that, especially for high mountain regions, both former habitat connectivity and historical climate-driven migration patterns may be reflected in the genetic structure of *Polylepis australis*. Comprehensive genetic studies may therefore represent very useful methods for understanding past forest dynamics, and for predicting future forest migration patterns.

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